Rapid Revision Summary

Last-minute review · condensed from 17 mark schemes and 16 papers, Jan 2019 – Jun 2026

THE PAPER

1 h 15 min · 70 marks · 6–7 context questions · ≈ 1 min per mark. Standard-form calculations, a method, a data evaluation and a 6-mark explanation. Whole spec including Paper-2-only content.

TOP 12 MARK-LOSERS

- Wrong standard form (18×10^{12}).
- "Amount", "food", "nutrients" — name the substance.
- Word equation when the symbol equation is asked.
- Codon/anticodon reversed.
- "Both genes expressed" for codominance.
- Phagocytes making antibodies; lymphocytes engulfing.
- "Fight/attack disease"; "immune" bacteria.
- Natural selection missing a step.
- "Traps light" for the greenhouse effect.
- "Nitrogen/phosphorus" instead of nitrate/phosphate.
- "Restricted enzyme", "lipase" for ligase; meiosis for stem cells.
- Generic answers that ignore the context.

INVESTIGATIONS

METHOD CHECKLIST

Same organism/size · change one variable · measure with named apparatus · set time · repeat + mean · named controls.

- Catalase: more discs → more active sites → more E–S complexes → more O_2; measure initial rate (substrate used up).
- Decay: cold slows enzymes; vinegar denatures them.
- Food energy: mass of water × 4.2 × ΔT; heat lost → low result.
- Cloudiness/colour is subjective — use a quantitative method.
- Exclude anomalies from means (4.2 ÷ 3).

PHOTOSYNTHESIS

- $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$.
- Red/blue absorbed (high rate); green reflected (lowest); no filter highest.
- Bubble counts inaccurate: sizes vary, O_2 dissolves → use a syringe.
- Hydrogencarbonate: light → purple (CO_2 used); dark → yellow; dim → no change.
- Palisade: top, packed, many chloroplasts. Spongy: air spaces. Stomata mostly underneath.
- CO_2 is a raw material and can limit the rate.

TRANSPIRATION

POTOMETER

Cut under water → seal with petroleum jelly → dry leaves → bubble distance in set time → move lamp → control temperature, humidity, wind → reset, repeat.

- More light → stomata open → more loss (until fully open).
- Wind removes humid air → steeper gradient → more diffusion.
- Oil on the water stops evaporation.
- 2 stomata in 0.0016 cm^2 = 1250/cm^2 × 150 = 187 500.
- Phloem: sucrose + amino acids.

EXCRETION AND HOMEOSTASIS

ADH (5)

Sweat → blood concentrated → hypothalamus → pituitary → ADH → collecting duct permeable → more water reabsorbed → less, concentrated urine.

- Dialysis: no urea in fluid, long coiled tube, thin membrane, counter-flow, 40 °C.
- Transplant: permanent, normal life; tissue match; immunosuppressants.
- Homeostasis: constant internal environment.

REPRODUCTIVE HORMONES

Hormone	Role
FSH (pituitary)	matures follicle; oestrogen
LH (pituitary)	ovulation; corpus luteum
oestrogen (ovary)	repairs lining; LH
progesterone	maintains lining; inhibits FSH, LH

- (51 − 12) × 13 ≈ 510 eggs.
- Low FSH → no egg maturation; low LH → no ovulation.
- Contraceptive evaluation: effectiveness, side effects, reversibility, STIs.

GENETICS

PROTEIN SYNTHESIS

DNA → mRNA (nucleus) → ribosome → tRNA brings amino acid → **anticodon binds codon** → amino acids join → polypeptide.

- Codominance: both **alleles** expressed (I^AI^B). O: no antigens, universal donor.
- Sickle cell: less SA → less O_2; carriers survive malaria.
- Polygenic (height) vs monogenic (hair colour).
- Two carriers → P(affected) = 0.25.
- Gene editing: off-target binding; ethics; no consent.

SELECTION AND BREEDING

NATURAL SELECTION

Mutation → variation → survive → reproduce → pass on allele → more common.

- Selective breeding: select best → breed → select offspring → repeat.
- Antibiotic resistance: same chain (MRSA).

IMMUNITY

- Phagocytes engulf; lymphocytes make antibodies + memory cells.
- Vaccine: attenuated/dead pathogen → antigens → antibodies + memory cells → faster, bigger response.
- Virus: protein coat, DNA or RNA, no cell structures.
- Mosquitoes: warm, humid, standing water.
- Pasteurisation kills bacteria.

ECOLOGY AND POLLUTION

EUTROPHICATION (4)

Fertiliser/sewage → algal bloom → blocks light → plants die → decomposers respire → O_2 falls → fish die.

NITROGEN CYCLE

Fixing: N_2 → ammonium/nitrate · decomposers: protein → ammonium · nitrifying: ammonium → nitrite → nitrate · denitrifying: nitrate → N_2.

- Greenhouse: traps infrared → global warming → ice melt, sea level, habitat loss.
- Deforestation: erosion, less transpiration, less CO_2 absorbed.
- Quadrats: random coordinates, repeat, mean.

BIOTECHNOLOGY

- Restriction enzyme cuts (sticky ends) → ligase joins → vector.
- Cloning: many, fast, identical, any time of year.
- Fermenter: paddles mix nutrients/O_2; water jacket cools; aseptic.
- Fish farm: filtered water, O_2, less disease, nets.
- Stem cells: undifferentiated, mitosis, differentiate.

CALCULATIONS

Type	Example
%	43 ÷ 1300 × 100 = 3.3%
% change	(8.3 − 7.3) ÷ 7.3 = 13.7%
standard form	9.5×10^8
ratio	147 710 : 13 297 = 11.1 : 1
rate	10 kg ÷ 180 d = 0.056 kg/d
scaling	14.6 × 60 × 24 = 21 024